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Beyond Seismic and Logs: Real-Time Geological Model Updates from UDAR-Based Inversions

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Abstract

Field development in thin, heterogeneous reservoirs presents persistent challenges due to geological complexities such as pinch-outs, shale interbeds, and abrupt changes in structural dip. These features are often below seismic resolution and difficult to capture using traditional 1D petrophysical logs alone. Consequently, predicting lateral reservoir continuity and sand geometry with sufficient confidence remains a significant uncertainty in well planning and reservoir modeling. Ultra-Deep Azimuthal Resistivity (UDAR) inversion provides a transformative solution by enabling high-resolution structural and stratigraphic interpretation around the wellbore.

This study explores the integration of UDAR inversion data to update geological facies models and improve reservoir characterization. UDAR's 2D and 3D inversion products, derived from advanced reservoir mapping while drilling (RMWD), offer insight into resistivity-based geo-bodies that indicate the presence and distribution of sand. These interpretations, when integrated with borehole image logs and traditional LWD measurements, support the development of more geologically realistic models in near real-time.

To demonstrate the value of this approach, a blind well test was performed using two sector models. The first was built using vertical/semi-vertical legacy well data only, while the second incorporated both vertical/semi-vertical wells and UDAR-derived sand body interpretations from multiple horizontal wells in the area. The second model showed a significantly improved match to the blind well's actual log responses, validating the contribution of UDAR interpretation in reducing geological uncertainty.

By capturing sub-seismic scale depositional features and delivering 360-degree structural updates around the wellbore, UDAR-based resistivity inversions enhance the spatial resolution of facies architecture. This approach revealed key depositional trends and stratigraphic variations across the study area, which are critical for optimizing reservoir performance and well placement.

The results underscore that 1D solutions are no longer sufficient in complex field environments. Integrating multi-dimensional UDAR inversion data into the geological modeling workflow provides a

more reliable foundation for reserves estimation and strategic field development, ultimately enabling more informed decisions in both near-field and full-field development planning.

Introduction

The development of hydrocarbon fields in heterogeneous reservoirs presents significant challenges due to the inherent structural and stratigraphic complexities. Such complexities include lateral variations in facies distribution, fault networks, and the presence of thin reservoir targets. These geological features frequently occur at scales that fall below the resolution limit of conventional seismic data, thereby impeding accurate characterization and modeling of reservoir architecture (Jin et al., 2018; Smith & Doe, 2020). Consequently, reservoir models constructed from traditional data often lack sufficient detail to capture the true spatial heterogeneity, which critically impacts well placement and production forecasting. Effective well planning and geosteering operations necessitate geological data of a resolution exceeding that typically provided by conventional modeling techniques. The common reliance on one-dimensional (1D) petrophysical evaluations, coupled with legacy well data as the primary field control, is often inadequate in complex reservoirs where lateral heterogeneities dominate. This inadequacy leads to uncertainties in delineating facies boundaries and fluid contacts, increasing the risks associated with drilling operations and reducing overall reservoir development efficiency.

Recent technological advancements in downhole measurement tools have introduced Ultra Deep Azimuthal Resistivity (UDAR) inversion as a transformative approach to subsurface imaging. UDAR technology enables the acquisition of high-resolution resistivity data at depths significantly beyond the wellbore, facilitating real-time imaging of resistive geological bodies and structural boundaries. The integration of UDAR inversion results with ongoing geosteering activities and three-dimensional (3D) geological modeling workflows allows for continuous updates to the reservoir model, thereby enhancing the spatial resolution of subsurface interpretations. In this study, we demonstrate the application of UDAR inversion data within a geosteering framework to refine facies models and reduce geological uncertainty. The workflow incorporates real-time resistivity inversion results into reservoir modeling software, enabling dynamic adjustment of facies distribution based on updated resistivity signatures. Comparative analyses with legacy workflows illustrate the improvements in model accuracy and well placement outcomes. The findings of this research underscore the substantial benefits of applying Ultra Deep Azimuthal Resistivity inversion technology in complex heterogeneous reservoirs. By surpassing the resolution limitations of conventional data, UDAR facilitates more precise reservoir characterization and informed decision-making during field development. The successful implementation of this technology and the associated integrated workflows demonstrate a paradigm shift in reservoir modeling, promoting more efficient and risk-mitigated well placement strategies.

Well planning and Geosteering activities requires higher resolution of the geological data that exceeds what conventional models can provide. Traditional 1D petrophysical evaluation and legacy of the well in the field as control, are insufficient in such environments. In consequence, reservoir models most of the time fail to capture the true spatial distribution of facies and fluid contacts, leading to poor placement results and increased drilling risks.

To address these limitations, the Ultra Deep Azimuthal Resistivity (UDAR) inversion technology provides a game-changing in subsurface imaging and it can be an adapted solution to enhance reservoir understanding and modelling in real time. By integrating UDAR inversions with geosteering operations and 3D geological modeling, UDAR enables the detection of resistive bodies and boundaries well beyond the wellbore. This allows for more informed decisions while drilling and significantly improves the geological models.

This paper presents a case of study demonstrating how UDAR data is used to update the facies models, reduce uncertainty and improve field development planning. The results show clear advantages over legacy workflows, highlighting the value of integrating UDAR data and the collaboration among geoscience teams.

Ultra-Deep Azimuthal Resistivity Mapping Technology

Ultra-Deep Azimuthal Resistivity (UDAR) has become essential in modern reservoir mapping-while-drilling operations, providing insights for geosteering, well placement, and real-time reservoir characterization. UDAR tools represent an advancement in directional resistivity logging, designed to provide detailed 3D images of the subsurface while drilling. This capability allows operators to map reservoir layers and fluid boundaries beyond the immediate drilling area with high resolution.

Resistivity logging technology has evolved significantly. Early tools measured resistivity with limited depth of detection around the borehole and assumed uniform surrounding rock, limiting their ability to detect complex features. Later, azimuthal resistivity tools measured resistivity in different directions around the borehole, helping identify dipping beds and fractures, but with limited depth and resolution. The breakthrough came with full electromagnetic (EM) tensor measurements. Modern UDAR tools use multiple transmitters and receivers to capture the full EM field, including magnitude, phase, and direction. This dataset allows detailed imaging of reservoir structure and fluid contacts. Inversion techniques have evolved from simple 1D models to advanced multi-dimensional algorithms, creating detailed 3D resistivity models matching measured data. Cloud computing now enables these inversions in real time. Figure 1 shows an example of 3D Reservoir Mapping While Drilling (RMWD) where geobody extraction from inverted resistivity around the wellbore helps understand the reservoir (Y. Abbas et al, 2024).

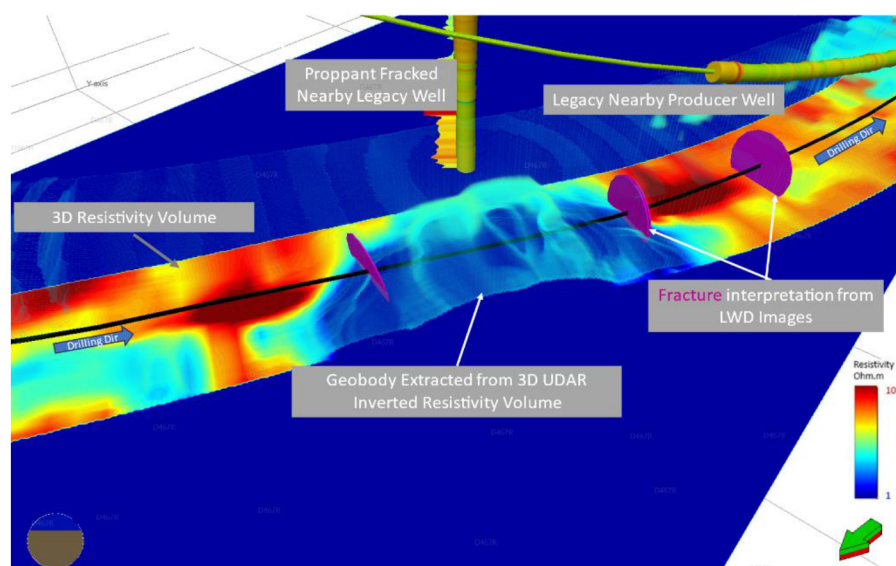


Figure 1—Example of latest advancement in multi-dimensional reservoir mapping with Ultra-deep Azimuthal resistivity (UDAR) (Y. Abbas et al, 2024)

UDAR hardware is built around modular transmitters and multifrequency receivers integrated into the bottomhole assembly, capable of probing tens to hundreds of feet into the formation—well beyond the reach of conventional logging-while-drilling technologies (Beer et al., 2010; Netto et al., 2012). This extended depth of investigation reduces reliance on pilot wells and supports proactive steering by imaging structure and fluid distributions in real time. A typical system includes one transmitter and up to three receiver modules distributed along the assembly (Figure 2). Each receiver measures at multiple frequencies—commonly 2, 6, 12, 24, 48, and 96 kHz—allowing configuration flexibility and improved signal-to-noise ratios across varying geological settings (Seydoux et al., 2014).

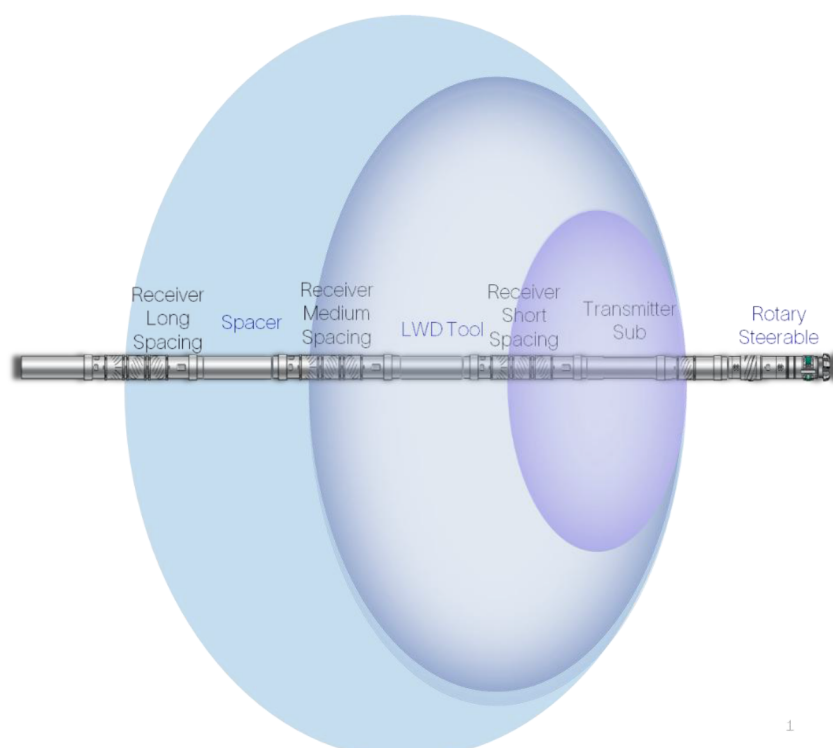


Figure 2—A borehole assembly with a UDAR transmitter and 2 UDAR receivers showing volumetric signal sensitivity to the formation surrounding the wellbore

During data acquisition, the UDAR tool records the complete resistivity tensor, generating a detailed, directionally sensitive image of the surrounding formation. These measurements capture both anisotropic behavior and azimuthal resistivity variations, which are essential for characterizing complex reservoirs. With receiver modules spaced up to 100 feet apart, the tool can investigate formation features at comparable radial distances from the borehole, though the effective depth depends on formation resistivity and the chosen configuration. Operators can select three optimal operating frequencies from six available, based on anticipated formation properties and pre-job planning. Figure 3 shows a representative UDAR tool assembly deployed across all wells in this study.

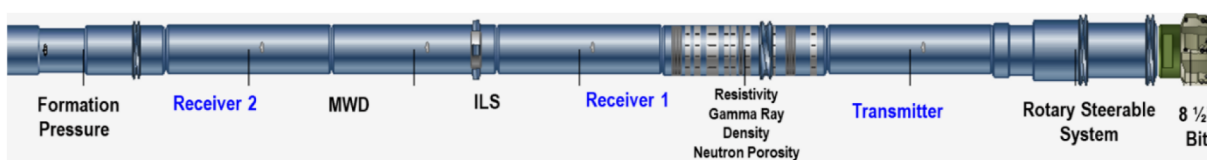


Figure 3—A borehole assembly with a UDAR transmitter and 2 UDAR receivers deployed for wells discussed in this paper.

Data processing begins with filtering, corrections for borehole effects, and tool-position adjustments. Inversion workflows are then applied. Early implementations used 1D deterministic and stochastic approaches that assumed lateral homogeneity (Dong et al., 2015; Wilding-Steele et al., 2019). These produced 2D curtain sections along the well path at fixed intervals, sufficient for simple formations but inadequate for complex geology. Transitioning to 2D inversion significantly improved interpretation by resolving lateral resistivity variations and capturing transverse structures (Thiel and Omeragic, 2018, 2019). By inverting components of the full EM tensor, modern algorithms recover horizontal and vertical resistivity and the angular relationship of bedding relative to the wellbore, enabling improved mapping of thin beds, shale barriers, and laterally varying fluid contacts.

Building on this, the latest UDAR developments have introduced full 3D interpretation workflows. Instead of limiting results to 2D curtain sections, these workflows integrate multi-dimensional inversion outputs with pixel-based interpretation methods and cloud-enabled processing platforms. The result is a volumetric representation of resistivity distributions around the wellbore, where adaptive regularization helps stabilize solutions and preserve resolution in high-contrast or structurally complex settings (Thiel et al., 2021). Validation against synthetic geological scenarios—including displaced sand bodies, shale intrusions, and multifingered injectites—has confirmed that 3D interpretation can reliably capture orientation, size, and resistivity contrasts of subsurface features (Wang et al., 2021). With cloud-based platforms, compressed datasets are transmitted in real time and distributed across scalable computing resources, enabling fast, automated inversions essential for high-angle and horizontal well geosteering. Depending on reservoir complexity and operational objectives, stochastic, deterministic, and pixel-based inversions can be run in parallel to maximize interpretive robustness. Stochastic approaches provide probabilistic ranges in noisy environments, while deterministic solutions deliver higher resolution for thin stratigraphic intervals (Wilding-Steele et al., 2019).

UDAR interpretation frameworks combine stochastic, deterministic, and pixel-based inversions to balance uncertainty quantification, resolution, and robustness (Dong et al., 2015; Wilding-Steele et al., 2019). Measurements from short- and long-spacing receivers are jointly inverted, with the former capturing near-borehole detail and the latter extending penetration depth. Together, these multi-scale inversions enhance both vertical and lateral imaging. High-definition UDAR configurations—equipped with more powerful transmitters and increased spacing up to 40 m—extend depth of investigation and further improve signal quality (Xu et al., 2018).

By linking logging-while-drilling data with high-resolution, real-time inversion models, UDAR technology bridges the gap between local measurements and field-scale reservoir models. Its ability to deliver 3D imaging of complex geology during drilling empowers operators to optimize well placement, reduce drilling risk, and improve hydrocarbon recovery.

Methodology

The study area is focused on a complex fluvial-deltaic system with channelized sand bodies interbedded within shale. These sands show significant lateral and vertical variability, making well planning and reservoir modeling challenging due to their sub-seismic resolution and unpredictable geometries.

In this particular case, the drilling campaign targeted three distinct stratigraphic layers within the same reservoir unit. The fluvial channels display abrupt pinch-outs and variable thicknesses, often eluding seismic detection and conventional mapping. As a result, real-time data from the Ultra-Deep Azimuthal Resistivity (UDAR) tool played a critical role in detecting these sand bodies during drilling operations.

The primary data set comprises several vertical legacy wells with comprehensive formation evaluation logs—including gamma ray, resistivity, density, and neutron logs and multiple horizontal wells drilled from a star-shaped platform using the UDAR tool (Fig. 4). These horizontal wells provided continuous, high-resolution azimuthal resistivity measurements that enabled detailed mapping of resistivity distribution around the borehole, illuminating sand body geometries otherwise invisible in the seismic volume (Fig. 5).

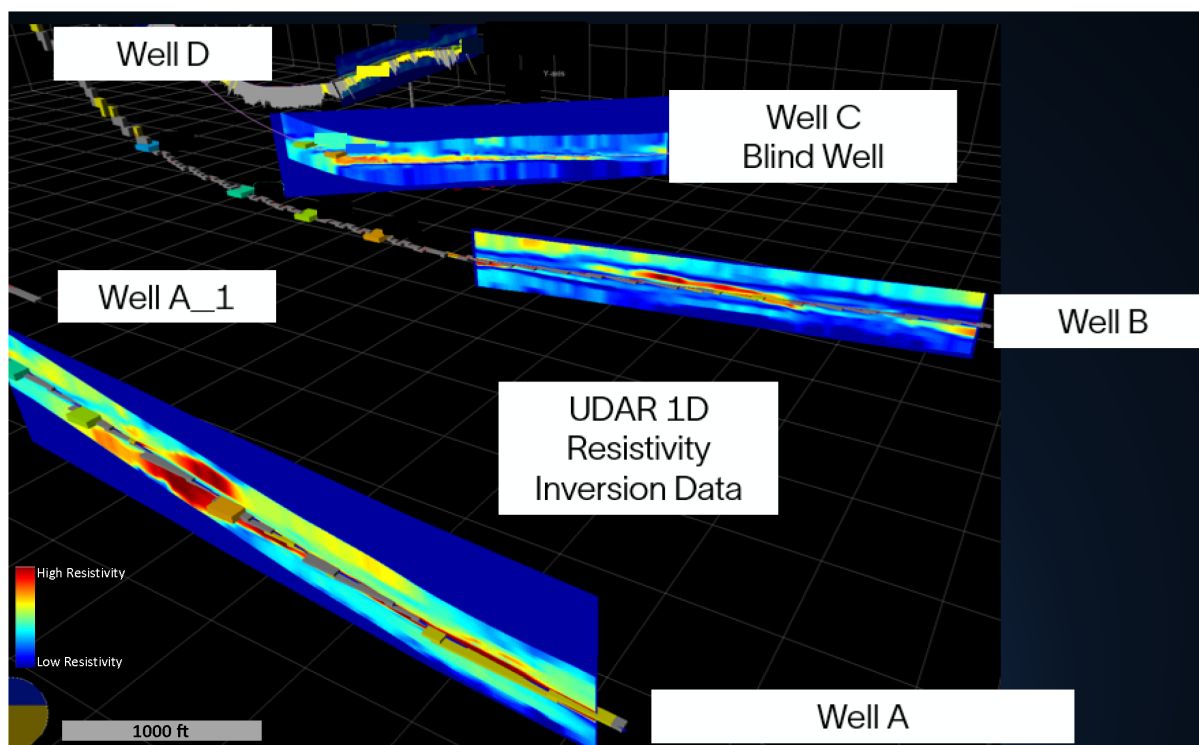


Figure 4—Star shaped drilling program with multiple horizontal wells using UDAR.

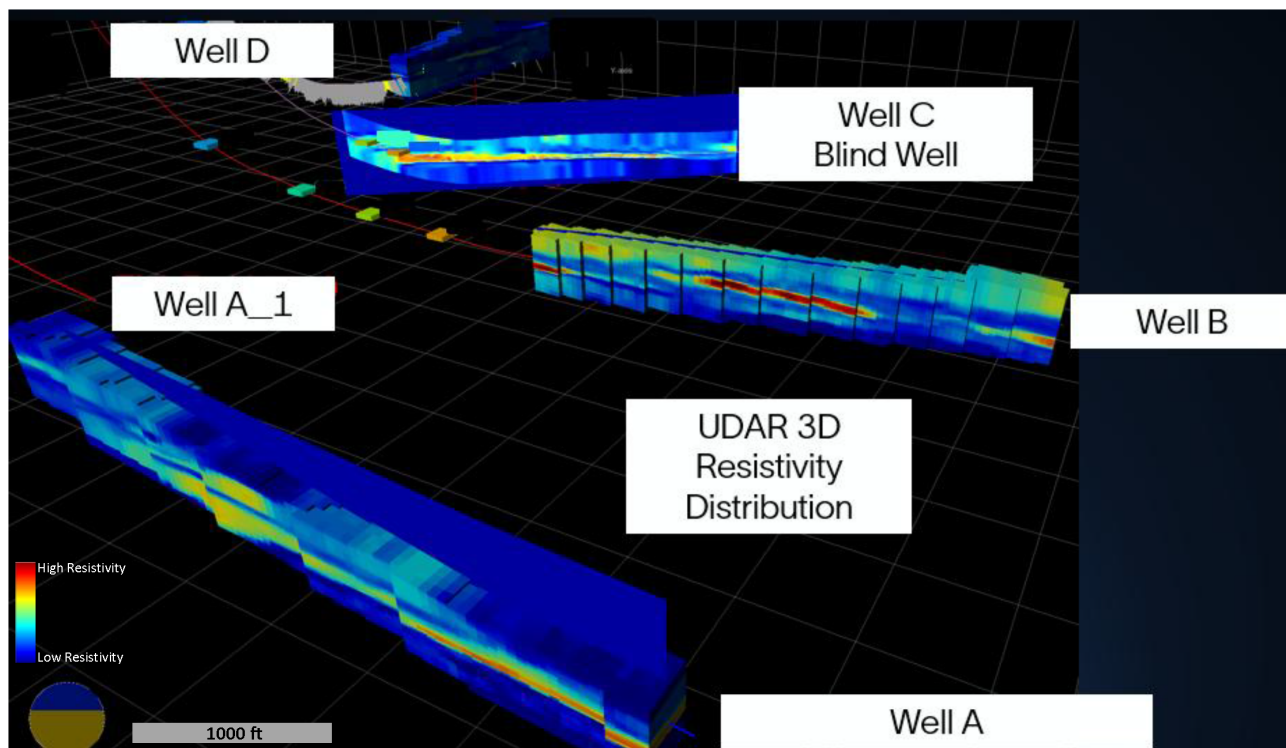


Figure 5—3D Resistivity distribution around the wellbore mapped by multi-dimensional UDAR

Integration of Inversion Results into the 3D Geological Model

A two-pronged facies classification approach was employed. The first method applied a standard log-based workflow applied to both legacy vertical and horizontal wells, using Gr, resistivity, density, and neutron logs to distinguish between sand and shale. The facies scheme was kept simple to focus on the gross reservoir

architecture and ensure consistency across data types. No further subdivision into levee, fine-grained, or coarse-grained sands was undertaken, in order to maintain consistency across data types.

The second classification used UDAR inversion results. High-resolution azimuthal resistivity profiles from horizontal wells were used to delineate sand geobodies based on resistivity thresholds. These thresholds were calibrated against legacy well data to ensure consistency with the petrophysical definition of sand. The UDAR-derived geobodies provided a novel representation of reservoir heterogeneity, capturing lateral sand continuity and pinch-outs undetectable in conventional data sets. (Fig. 6). The resulting output from these two workflows comprised a traditional facies map based on formation evaluation logs and a UDAR-enhanced facies map delineating resistive sand bodies distributed around the wellbore in three dimensions.

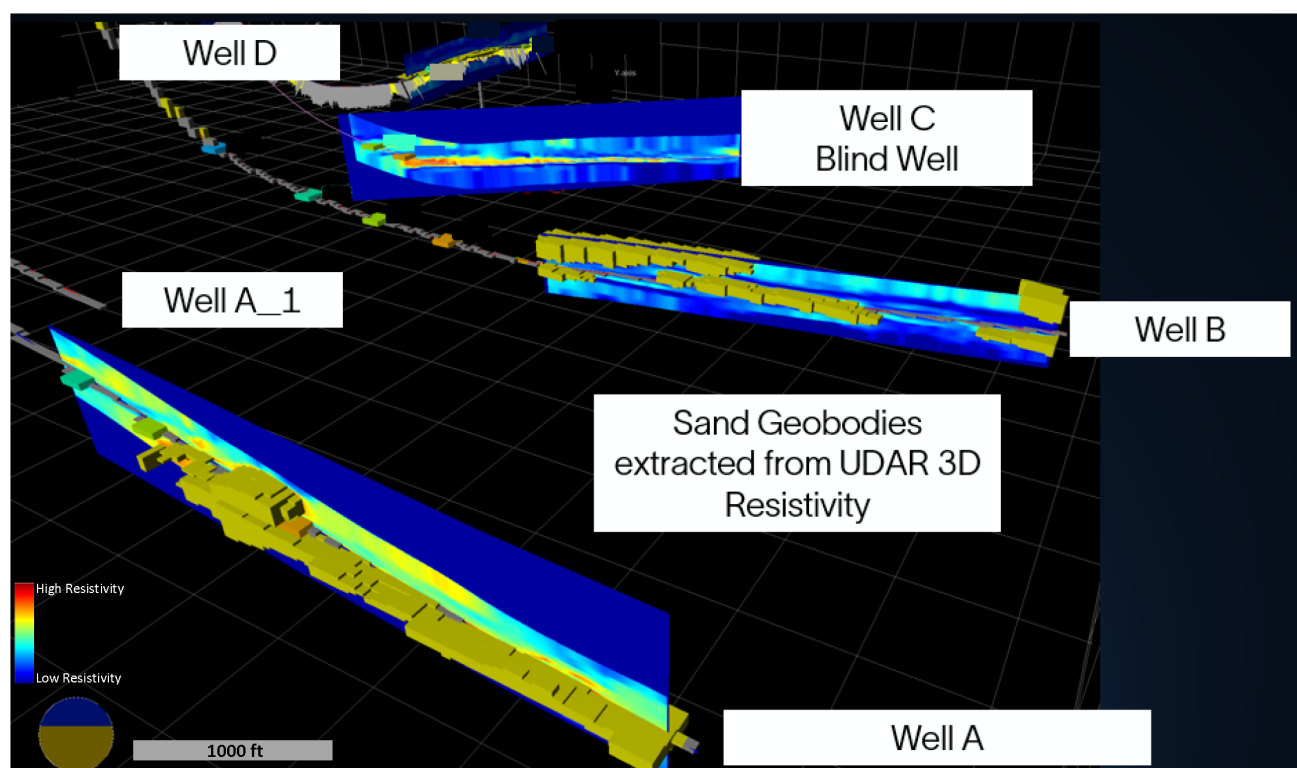


Figure 6—Geobodies extracted from 3D UDAR data after classification of facies using pre-established resistivity thresholds linked to lithology (as a function of Gamma Ray, Resistivity, Density/Neutron Porosity).

To integrate facies classification into 3D geological model, a Sequential Indicator Simulation (SIS) algorithm was used. SIS is a commonly used geostatistical technique for modeling categorical or discrete variables, such as lithofacies. This method is especially suitable for modeling heterogeneous fluvial reservoirs, where facies distributions are complex and non-Gaussian. It allows for sharp transitions between facies types and preserves geological realism, which is often smoothed out in Gaussian-based approaches. The implementation of SIS in this study involved the integration of both hard data (well log-derived facies interpretations and core data) and soft data (facies probabilities inferred from resistivity inversion models and other indirect measurements).

Variograms, which quantify the spatial continuity and correlation structure of geological variables were constructed using the interpreted facies logs and distributions derived from UDAR (Unsupervised Diffusion-based Attribute Recognition), a method used to extract sand-body trends from geophysical logs or seismic volumes. These variograms were fitted with theoretical models, typically spherical or exponential, depending on the nature of the spatial correlation observed in the data. The chosen model parameters;

including nugget, sill, and range were then input into the SIS algorithm to guide facies distribution simulation.

In this study, two distinct facies modeling approaches were developed and evaluated for their effectiveness in representing subsurface heterogeneity:

Baseline Facies Model: This model was constructed exclusively using standard log-derived facies data obtained from both legacy vertical wells and horizontal wells. The interpretation relied primarily on conventional well log measurements, including gamma ray, resistivity, and density logs, to delineate lithofacies.

Enhanced UDAR-Integrated Facies Model: In contrast to the baseline model, the enhanced model incorporated additional geological insights derived from Ultra-Deep Azimuthal Resistivity (UDAR) inversion outputs. Specifically, it integrated both the traditional log-derived facies and resistivity-based geobodies interpreted from UDAR data, which enabled the identification and delineation of sand-rich zones beyond the immediate wellbore environment (Katterbauer et. al., 2015).

The inclusion of UDAR-derived geobodies in the facies modeling improved the geological realism and spatial continuity of depositional elements, particularly sand bodies. This was especially beneficial in inter-well regions, where data is sparse and interpolation is uncertain. The UDAR-integrated model provided higher resolution in resolving sand body geometry and distribution, enhancing the delineation of lateral facies continuity and channel connectivity. This improved predictive capability is crucial for reservoir modeling and simulation, impacting hydrocarbon flow path estimation, reservoir compartment identification and overall recovery efficiency.

The findings underscore the importance of integrating advanced resistivity-based inversion technologies like UDAR into subsurface modeling workflows to reduce geological uncertainty and enhance reservoir characterization in complex settings.

Statistical evaluation of facies proportions, continuity, and distribution trends was conducted across both vertical and horizontal wells. Directional variograms were critical due to the anisotropic nature of fluvial deposition, constraining the modeling in directions parallel and perpendicular to paleoflow.

Two facies models were generated:

1. Baseline Model – constructed using only the standard log-derived facies from legacy and horizontal wells.
2. Enhanced UDAR-Integrated Model – built using both the standard log-derived facies and the resistivity-derived sand geobodies from UDAR inversions.

The integration of UDAR-derived data into the facies model provided a more realistic portrayal of sand body geometry, especially between wellbores where interpolation is uncertain. This enhanced model improved the prediction of lateral connectivity and channel continuity.

Model Comparison and Quality Control

To assess the impact of integrating Ultra-Deep Azimuthal Resistivity (UDAR) inversion results into the facies modeling workflow, a blind well test was performed within the studied platform. Among the multiple horizontal wells drilled from the star-shaped platform, one well—hereafter referred to as Blind Well C—was deliberately excluded from both the baseline and UDAR-enhanced facies modeling processes. Both its formation evaluation log data and its UDAR inversion results were withheld to provide an independent dataset for model validation.

Two complementary quality control (QC) techniques were used to evaluate the performance of the facies models. The first approach involved a qualitative visual comparison between the two facies models at a fixed true vertical depth corresponding to the depth of Blind Well C. Two depth-slice snapshots were generated, one from the baseline model and another from the UDAR-enhanced model (Fig. 7).

- The baseline model, based on offset well data and conventional petrophysical logs, shows a significant mismatch with the UDAR inversion results around Blind Well C. While UDAR data indicates sand bodies above, below, and along the wellbore, the baseline model predicts a dominantly shale environment.
- The UDAR-enhanced model, incorporating sand geobodies from UDAR inversion data, closely aligns with the actual sand distribution from Blind Well C's UDAR log. This model captures sand continuity and pinch-outs, highlighting the value of UDAR-derived resistivity in facies modeling.

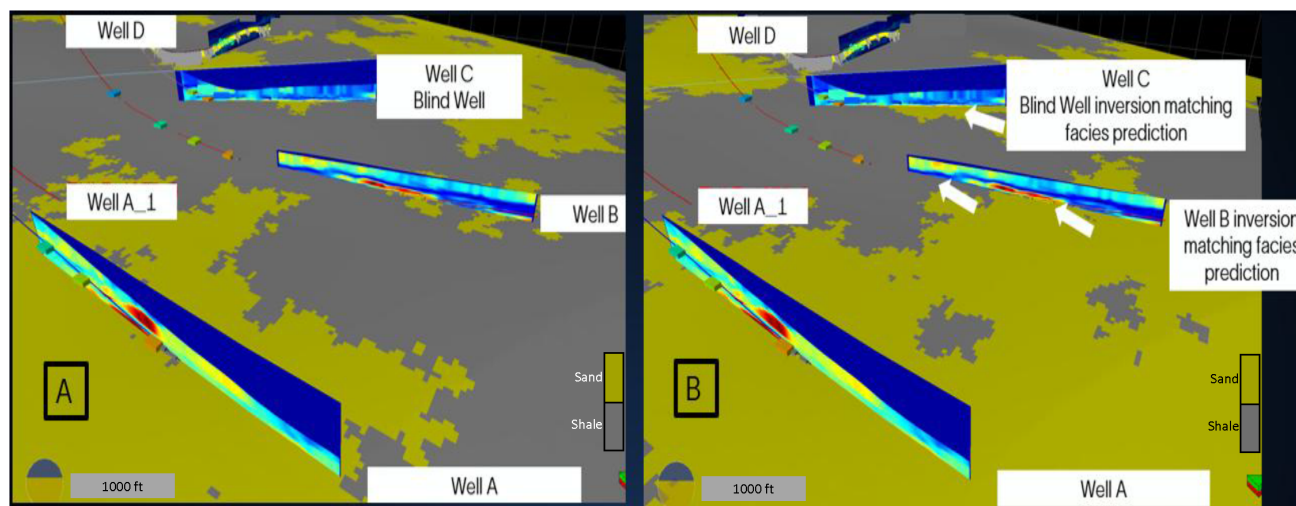


Figure 7—Visual QC of two facies model, A) Baseline Model, B) UDAR Enhanced Model

A second QC strategy quantitatively assessed the predictive accuracy of two facies models using independent well data. Predicted facies logs along Blind Well C's trajectory were compared with actual formation data (refer to Fig. 8). The models analyzed were: the baseline facies model, based on conventional log-based classification, and the enhanced facies model, incorporating UDAR-derived resistivity geobodies.

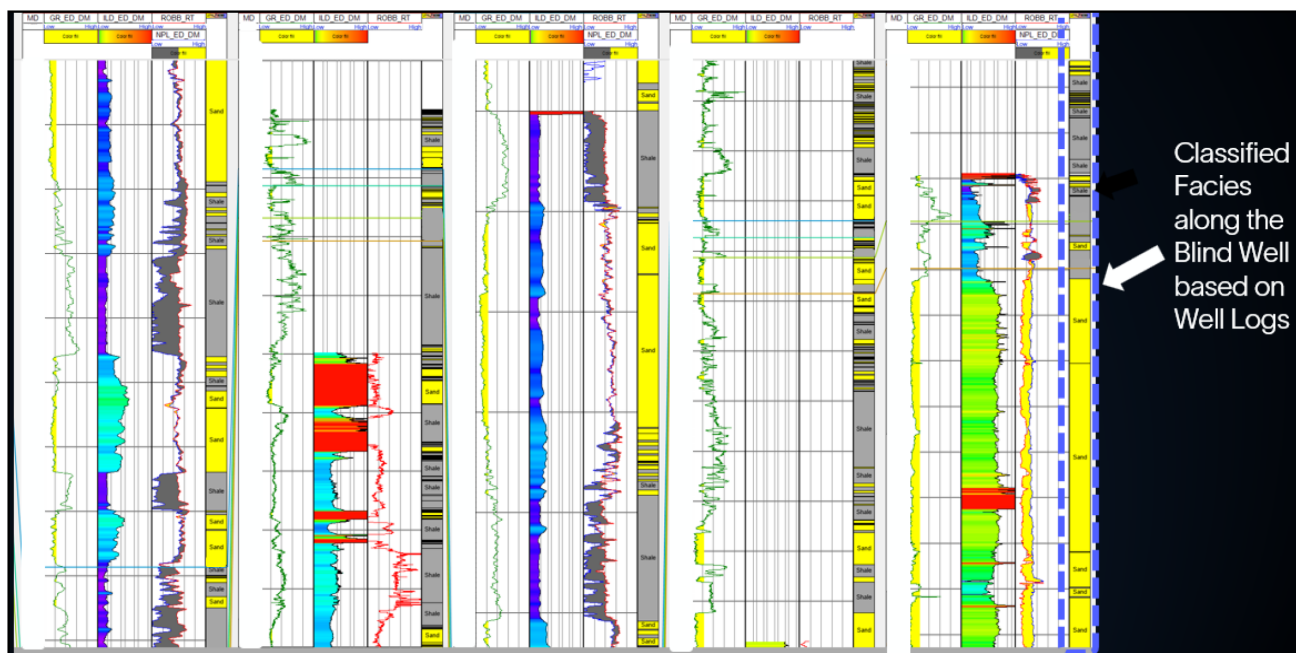


Figure 8—Standard Facies classification of the offset wells and the Blind well 'C'

The facies log generated from the baseline model exhibited substantial discrepancies when compared with the actual lithological profile of Blind Well C. Notably, extensive intervals predicted as shale in the model were, in reality, identified as sand in the well data. These misclassifications highlight the limitations of relying solely on conventional petrophysical logs and spatial interpolation from offset wells, particularly in geologically complex settings where lateral heterogeneity is pronounced.

In contrast, the UDAR-enhanced model showed improved alignment with observed lithologies, accurately capturing 80-85% of true sand and shale intervals identified in Blind Well C. This level of agreement suggests that the integration of resistivity geobodies derived from UDAR workflows, introduces valuable geological constraints that enhance the spatial resolution and predictive fidelity of facies distribution. The resulting model not only reduces uncertainty but also exhibits a more geologically consistent delineation of sand body geometries (see Fig. 9).

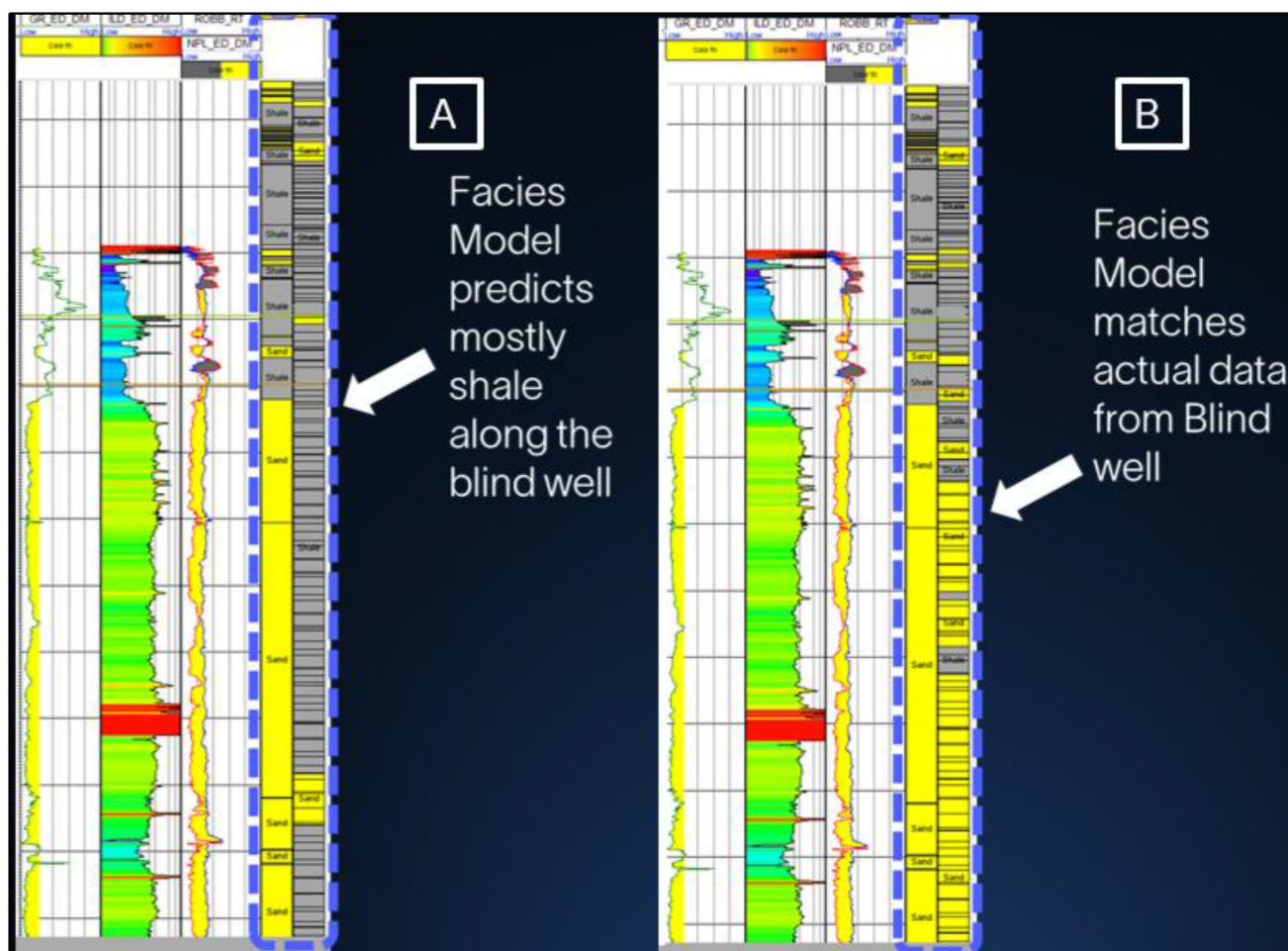


Figure 9—Quantitative comparison of the Facies data extracted from the two models along the Blind well ‘C’ (second facies track), A) Baseline Model, B) UDAR Enhanced Model

These findings affirm the value of advanced machine learning techniques; specifically, the UDAR approach - in augmenting traditional reservoir characterization workflows. By integrating data-driven resistivity anomalies as geological priors, the UDAR-enhanced model captures subsurface heterogeneity with greater precision, thus offering a more reliable framework for subsurface interpretation and decision-making. The second QC approach involved a quantitative comparison of extracted facies logs along the trajectory of Blind Well C (Fig. 8) from both models. Facies predictions along the wellbore path were extracted from: the baseline facies model, and the UDAR-enhanced facies model.

These extracted logs were plotted alongside the actual formation evaluation data from Blind Well C, serving as ground truth.

The results clearly indicate that:

- The baseline model shows significant deviation from the observed lithology. Large sections predicted as shale in the model correspond to actual sand intervals in the real well data, highlighting poor predictive performance when relying solely on offset wells and conventional log-based classification.
- The UDAR-enhanced model demonstrates a marked improvement in alignment with actual well data. The extracted facies log from this model matches approximately 80–85% of the true sand and shale intervals observed in Blind Well C. This alignment confirms that the inclusion of UDAR-derived resistivity geobodies provides a more accurate and geologically consistent prediction of sand body distribution (Fig. 9).

This comparison not only validates the efficacy of integrating UDAR inversion results into the facies modeling workflow but also demonstrates the limitations of traditional approaches in complex fluvial systems where sand body geometry is highly variable and below seismic resolution.

Beyond structure, UDAR inversions offer critical insights into facies distribution. As previously described, sand bodies can be interpreted from resistivity geobodies using calibrated thresholds developed from offset well analysis. The same criteria, resistivity cutoff, vertical extent, and lateral continuity can be applied in real time to classify sand vs. shale facies as the well progresses.

The real-time facies update consists of the following steps:

1. **Geobody Detection:** As the UDAR tool advances, new inversion data is processed continuously, and resistive geobodies are extracted around the borehole in near real time.
2. **Facies Assignment:** These geobodies are classified into facies using pre-established resistivity thresholds linked to lithology (as a function of Gamma Ray, Resistivity, Density/Neutron Porosity). Since the resistivity-facies relationship is calibrated offline, it can be deployed in real time with minimal computation.
3. **Model Conditioning:** The current 3D facies model is then updated to incorporate the new facies information. This can be done via:
 - Sequential Indicator Simulation (SIS) with soft conditioning, where the resistivity-derived facies are used as soft data to influence the next realization of the facies model. SIS is particularly suited to categorical facies classification, ensuring sharp boundaries between lithotypes and maintaining geological realism as the model evolves.
 - Bayesian updating techniques, where the prior facies probability (from the pre-drill model) is revised based on new UDAR evidence, thus updating the posterior probability distribution of facies presence.
 - Direct voxel replacement, where resistive geobodies are directly inserted into the model grid in the region of detection, assuming deterministic classification.

By continually feeding the model with high-resolution, spatially distributed resistivity-derived facies inputs, this approach maintains an up-to-date geological model that evolves alongside the drilling process. Importantly, this enables geosteering decisions not just based on structure, but on the evolving probability of encountering reservoir-quality facies.

Summary of Methodology

To summarize, the real-time model update framework leverages the capabilities of Ultra-Deep Azimuthal Resistivity to dynamically refine both structural and facies models while drilling. Structural boundaries

are interpreted and inserted into the model using seismic-style workflows, while resistivity geobodies are classified and integrated into the facies model using calibrated thresholds (function of GR, Resistivity & Density/Neutron Porosity) and geostatistical techniques. These innovations transform static geomodels into living models, continuously improving predictive accuracy and reducing geological uncertainty during reservoir navigation.

Simulation Study and Results

The practical efficacy of the workflow developed in this study was rigorously assessed through its application to a field development drilling campaign in a frontier hydrocarbon province. This development entailed the strategic placement of three horizontal well laterals, each targeting discrete stratigraphic horizons within a singular fluvial reservoir system. Although these horizons are encompassed within the same overall reservoir package, they are vertically separated by non-reservoir intervals that likely serve as regional seals, effectively precluding hydraulic communication between the layers.

Each lateral was purposefully aligned to penetrate a distinct stratigraphic horizon, facilitating a sequential drilling approach that enabled the progressive testing and iterative refinement of the integrated structural and facies modeling workflow. Central to this approach was the real-time assimilation of Ultra-Deep Azimuthal Resistivity (UDAR) inversion data, which provided dynamic updates to the structural framework. These structural updates were subsequently leveraged to inform refinements of the facies models, consistent with the methodology delineated earlier in the study.

The initial lateral drilling operation represented the first implementation of UDAR-based structural interpretation in a real-time setting. As the drilling progressed, continuous inversion results from the UDAR tool facilitated the delineation of the boundaries among the three principal reservoir layers. These real-time interpretations were systematically incorporated into structural grids, thereby enhancing the precision of depth estimations and the geometric characterization of reservoir units. This enhanced structural understanding during drilling was instrumental in enabling mid-drill trajectory corrections, thereby optimizing the wellbore placement within the target stratigraphy.

Concomitantly, resistivity-based facies classification was deployed in real time to provide preliminary updates to the facies model, particularly within the immediate vicinity of the borehole. These enriched facies framework served as a critical baseline for the planning and execution of the second lateral, fostering an improved geological context for subsequent drilling operations.

The planning of the second lateral benefited markedly from the updated structural and facies models derived during and subsequent to the first lateral's drilling phase. Integration of resistivity-defined boundaries and geobody information extracted from UDAR inversion data significantly refined the initial positioning of the second well. This advancement yielded several operational benefits, including:

- Enhanced anticipation of sand-shale interfaces, improving the resolution of lithological transitions.
- More accurate reservoir entry at the designated stratigraphic depth window.
- Reduced need for reactive trajectory modifications during geosteering, attributable to the improved fidelity of the geological model.

As drilling of the second lateral advanced, additional UDAR data continued to inform real-time adjustments, facilitating incremental refinements to the structural and facies models. These refinements cumulatively augmented operational confidence and geological certainty in preparation for subsequent wells.

By the time the third lateral was initiated, both the structural and facies models had undergone two iterative cycles of real-time updates, culminating in significantly enhanced reservoir characterization. The planning and execution of the third lateral capitalized on:

- A substantially more precise conceptualization of reservoir architecture.
- Improved prediction of reservoir entry points and the spatial configuration of sand bodies.

Empirically, the third well drilling operation demonstrated the heightened predictive accuracy afforded by the workflow, as evidenced by the minimal requirement for intervention during geosteering. The majority of trajectory adjustments were preemptively anticipated by the updated model, underscoring the robustness and operational value of the real-time integration of UDAR data.

The successful application of this workflow in a frontier setting underscores the transformative potential of integrating real-time geophysical inversion data into drilling operations. The utilization of UDAR inversion data offers a distinct advantage by enabling dynamic, data-driven updates to subsurface models, which is critical in complex fluvial depositional systems where heterogeneity and stratigraphic compartmentalization present significant challenges. Furthermore, the iterative refinement approach exemplified in this study highlights the value of adaptive modeling frameworks that evolve in response to incoming data, thereby reducing geological uncertainty and operational risk.

Beyond immediate drilling benefits, such methodologies contribute to more efficient reservoir development strategies, optimized well placement, and ultimately enhanced hydrocarbon recovery. Future research should explore the integration of additional data sources—such as real-time petrophysical logging, microseismic monitoring, and machine learning algorithms—to further enhance the predictive capability and automation of subsurface modeling workflows in analogous complex depositional environments. To evaluate the practical value of the workflow developed in this study, the approach was applied to a real drilling campaign in a frontier field (name withheld for confidentiality). The field development included the placement of three horizontal laterals, all targeting different stratigraphic layers of the same fluvial reservoir system. While these layers belong to the same reservoir package, they are separated by non-reservoir intervals—likely regional seals—and are not hydraulically connected.

Each lateral targeted a distinct stratigraphic horizon, and the sequence of drilling allowed the team to progressively test and refine the structural and facies modeling approach in a stepwise manner. The real-time integration of Ultra-Deep Azimuthal Resistivity (UDAR) inversion data enabled dynamic structural updates, followed by facies model refinements, as described in the methodology.

The first lateral provided the initial opportunity to apply UDAR-based structural interpretation in real time. As drilling progressed, inversion results from the UDAR tool helped delineate the boundaries of the three main reservoir layers. These real-time interpretations were used to update the structural grids, providing more accurate depth and geometry estimations of the reservoir units.

This real-time structural understanding helped refine the well trajectory mid-drill. In addition, the resistivity-based facies classification was also applied, providing early input for updating the facies model—especially around the borehole. This enriched model formed the baseline for planning the second lateral.

The second lateral was planned using the updated structure and facies model built during and after drilling the first well. Because the updated model already incorporated resistivity-defined boundaries and geobody information from the UDAR inversion, the starting position for the second well was significantly improved. This led to:

- Better anticipation of sand/shale transitions.
- Improved entry into the reservoir at the correct depth window.
- Fewer reactive changes during geosteering, since the expected geology was better matched.

As the second lateral progressed, further UDAR data allowed real-time adjustments to be applied again. This incremental refinement of the structural and facies models improved confidence for future operations.

By the time the third lateral was drilled, both the structural and facies models had been enhanced through two iterations of real-time UDAR updates. The planning of this third well benefitted from:

- A far more accurate understanding of reservoir architecture.
- Better prediction of reservoir entry points and expected sand body geometries.

In fact, the real-time model during the third well required minimal intervention during drilling. Most trajectory adjustments made during geosteering were already anticipated by the model, demonstrating a high degree of predictive accuracy.

Model Validation Against a Blind Vertical Pilot Well

To validate the UDAR-enhanced model, a nearby vertical pilot well (not used in the modeling process) was used for blind comparison. This pilot well provided a clean dataset of formation evaluation logs—including gamma ray, density, neutron, and resistivity—which could be compared directly with the model predictions.

When the facies distribution from the UDAR-enhanced model was extracted along the pilot well trajectory, the match with actual well data was remarkably close in the reservoir zone, both in terms of:

- Facies classification (sand vs. shale).
- Structural depth of reservoir tops and bases.

This strong match with an independent well further reinforced the validity of the real-time updated model.

In traditional workflows, UDAR data is often used only in a limited capacity: it helps steer the well during drilling but is not incorporated back into the geological model. As a result, even after acquiring high-quality UDAR data, the next well is typically planned, using the pre-existing static model built from offset vertical wells and legacy logs. This creates a disconnect between what was observed while drilling and what is used for planning.

In contrast, the workflow presented in this study closes that feedback loop. By interpreting UDAR inversion data in real time and updating both structure and facies models during the drilling process, the following benefits were demonstrated:

- Improved pre-job understanding of the subsurface ahead of bit.
- Reduced steering corrections due to better structural prediction.
- Better-informed decision-making for subsequent well planning.
- Stronger alignment between model predictions and actual geology encountered.

By the third lateral, the model had evolved to the point where its predictions required very few manual overrides a significant shift from the reactive and uncertain steering experienced during the first well.

Conclusions

Ultra-Deep Azimuthal Resistivity (UDAR) data represents a rich yet underexploited repository of geological information, with applications that extend significantly beyond its traditional use in geosteering operations. This study proposes a comprehensive, practical framework designed to systematically extract, classify, and integrate UDAR inversion outputs into both structural and facies modeling workflows. The methodology facilitates dynamic, real-time updates of subsurface models, thereby promoting a closer alignment between the interpreted geological framework and the actual conditions encountered in the subsurface environment.

- **Interpretation of UDAR Inversion Profiles:** UDAR inversion profiles can be analyzed in a manner analogous to seismic datasets, enabling real-time revisions of structural frameworks during drilling. This approach allows for more accurate delineation of faults, stratigraphic boundaries, and reservoir compartmentalization.
- **Resistivity-Based Facies Classification:** When calibrated against formation evaluation logs (such as gamma ray, density, and neutron porosity), resistivity data from UDAR inversions facilitate the consistent classification of facies. This process supports the robust integration of geobodies into three-dimensional (3D) reservoir models, improving the spatial characterization of heterogeneous lithologies.
- **Real-Time Model Updating:** The implementation of continuous, iterative model refinement using UDAR data significantly reduces geological uncertainty during drilling operations. Such updates enhance decision-making processes related to well-trajectory adjustments and optimize subsequent planning and development strategies.

The proposed workflow effectively closes the iterative loop between data acquisition, interpretation, and model refinement. By transitioning from static, legacy-based reservoir models to an agile, data-driven paradigm, the approach delivers substantial improvements in predictive accuracy and operational efficiency within reservoir development.

Despite these advancements, the integrity and reliability of UDAR data are often compromised by noise, measurement errors, and edge artifacts arising from sensor limitations and inversion processes. Edge artifacts, in particular, manifest as distortions near the boundaries of inversion zones, potentially leading to misinterpretations of subsurface features. To address these challenges, artificial intelligence (AI) methodologies offer promising solutions that can be integrated into the UDAR data processing workflow to enhance data quality control (QC) and artifact mitigation.

Machine learning (ML) algorithms, including supervised and unsupervised models, can be trained on labeled datasets containing examples of both high-quality and compromised UDAR data. These models can automatically detect anomalous readings, inconsistencies, or noise patterns that would otherwise require manual QC efforts. For instance, convolutional neural networks (CNNs) can analyze spatial patterns in resistivity profiles to identify and flag questionable data segments, facilitating rapid and objective QC processes.

AI techniques, such as deep learning-based image segmentation and denoising autoencoders, can effectively isolate and suppress edge artifacts in UDAR inversion results. By learning characteristic features of true geological signals versus artifacts, these models can enhance the fidelity of inversion outputs, particularly at boundary zones. Moreover, generative adversarial networks (GANs) have been employed in related geophysical domains to reconstruct missing or corrupted data, offering a potential avenue for artifact correction in UDAR datasets.

Incorporating AI-based QC and artifact removal algorithms into real-time data processing pipelines enables continuous data validation and refinement during drilling operations. This integration not only improves the accuracy of structural and facies interpretations derived from UDAR data but also supports more reliable model updates, further reducing geological uncertainty and operational risks.

The integration of UDAR data into reservoir modeling propriety workflows, augmented by AI-enabled data quality control and artifact removal, represents a significant advancement in subsurface characterization. This approach enables more accurate, timely, and reliable updates to geological models, fostering a shift toward more agile and data-driven reservoir development practices. Future research should focus on developing robust AI models tailored to the unique characteristics of UDAR data and validating their performance in diverse geological settings.

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